

REVIEW

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Probiotics and gastrointestinal disorders: an umbrella meta-analysis of therapeutic efficacy

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Abstract

Background Probiotic interventions are increasingly considered for alleviating gastrointestinal disorders such as diarrhea, nausea, bloating, and epigastric discomfort. This umbrella meta-analysis aims to resolve inconsistencies in existing research and offers a comprehensive assessment of probiotics' impact on these gastrointestinal symptoms.

Methods We conducted an extensive literature search across PubMed, Scopus, Web of Science, and Google Scholar (up to June 2024), including meta-analyses of interventional studies investigating probiotics' effects on gastrointestinal disorders. Pooled effect sizes with 95% confidence intervals were derived using a random-effects model, accompanied by subgroup and sensitivity analyses. The certainty of evidence and study quality were evaluated using AMSTAR 2 tools.

Results Probiotic supplementation was associated with a significant reduction in the risk of diarrhea (RR 0.44; 95% CI 0.37–0.52), nausea (RR 0.59; 95% CI 0.49–0.60), epigastric pain (RR 0.71; 95% CI 0.56–0.87), bloating (RR 0.74; 95% CI 0.64–0.84), and taste disturbance (RR 0.55; 95% CI 0.36–0.75), all with p-values < 0.001. Subgroup analyses revealed more pronounced effects in studies with shorter intervention durations (≤ 2 –4 weeks), and multi-strain formulations, particularly for diarrhea and epigastric pain. However, moderate to high heterogeneity and generally low methodological quality among several included meta-analyses limit the robustness of the findings.

Conclusion This umbrella meta-analysis demonstrates the potential effectiveness of probiotics in alleviating gastrointestinal symptoms. However, the findings should be interpreted with caution due to moderate to high heterogeneity and the generally low methodological quality of many included studies.

Keywords Probiotics, Diarrhea, Nausea, Gastrointestinal disorders, Umbrella meta-analysis

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Introduction

Gastrointestinal (GI) disorders present with a variety of symptoms, including diarrhea, nausea, bloating, and epigastric discomfort [1], which are frequently associated with conditions such as irritable bowel syndrome (IBS), inflammatory bowel disease (IBD), gastroesophageal reflux disease (GERD), and gastrointestinal infections [2]. These symptoms can arise due to disruptions in gastrointestinal function, inflammation, altered motility, or microbial imbalances. In particular, nausea is commonly linked to GERD [3], gastritis, excessive alcohol consumption [4], and the use of nonsteroidal anti-inflammatory drugs [5], while bloating often occurs in IBS, small intestinal bacterial overgrowth (SIBO), and functional constipation [6–8]. When persistent, symptoms such as diarrhea and nausea may lead to dehydration [9] and impair nutrient absorption, contributing to unintended weight loss, fatigue, and deficiencies in essential vitamins and minerals [10]. Moreover, chronic fluid and electrolyte imbalances associated with these symptoms can result in complications such as muscle weakness, cardiac arrhythmias, or, in severe cases, seizures [11]. The cumulative burden of GI symptoms often leads to reduced physical performance [12], diminished social functioning, and psychological distress, thereby impairing overall quality of life [13]. In light of these consequences, identifying effective supportive strategies—particularly nutritional interventions—has become a key focus in the management of gastrointestinal disorders.

Probiotics are live microorganisms that, when administered in adequate amounts, confer health benefits to the host [14, 15]. They have long been considered a supportive treatment for various gastrointestinal conditions [16], particularly in preventing or alleviating antibiotic-associated diarrhea (AAD) [17]. Probiotics have also been studied for enteric feeding diarrhea [18], *Helicobacter pylori* gastroenteritis [19], sucrose-maltase deficiency [20], IBD [21], SIBO [22], and lactose intolerance [23]. Research studies have demonstrated that probiotics can impede the activity of gut bacterial enzymes that play a crucial role in producing colonic carcinogens [24]. Probiotics exert several strategies to promote intestinal health, encompassing immune stimulation, nutritional competition, suppression of epithelial and mucosal adhesion, prevention of epithelial invasion, and synthesis of antimicrobial compounds [25].

Numerous meta-analyses have examined the impact of probiotic supplementation on the risk of gastrointestinal disorders—particularly diarrhea and nausea—across various conditions, including IBS and *Helicobacter pylori* infections [26–47]. However, the findings have been inconsistent, with several studies reporting significant heterogeneity. For example, concerning diarrhea

outcomes, four studies have yielded findings that lack statistical significance [32, 33, 37, 39], while other studies have shown significant findings [34–36, 38, 42, 43, 47, 48]. Given these discrepancies, the present umbrella meta-analysis aims to provide a comprehensive evaluation of the effects of probiotics on gastrointestinal disorders, including diarrhea, nausea, bloating, epigastric pain, taste disturbances.

Methods

This investigation was executed according to the standards of the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) to ensure a systematic and rigorous methodology [49]. An extensive search was performed in significant international scientific databases, notably PubMed, Scopus, Web of Science, and Google Scholar to locate articles of interest. The search includes all articles available in each database from its establishment up to June 2024. The search strategy was built based on relevant keywords as follows: [(Probiotic[Title/Abstract] OR probiotics[Title/Abstract]) AND (“Gastrointestinal complications” [Title/Abstract] OR “Gastrointestinal side-effects” [Title/Abstract] OR diarrhea[Title/Abstract] OR nausea[Title/Abstract] OR bloating[Title/Abstract] OR taste disturbance[Title/Abstract] OR epigastric pain[Title/Abstract])] AND (“Meta-analysis”[Title/Abstract] OR “meta analysis”[Title/Abstract]).

Inclusion and exclusion criteria

The PICO criteria for this umbrella meta-analysis were as follows: Population (P: adults), Intervention (I: probiotics), Comparison (C: control or placebo group), Outcome (O: gastrointestinal disorders including diarrhea, nausea, bloating, taste disturbance, epigastric pain). This umbrella meta-analysis included only meta-analyses of randomized controlled trials (RCTs) to ensure robust evidence by minimizing biases such as confounding and selection bias. Observational and quasi-experimental studies, as well as in vitro, in vivo, and ex vivo research, were excluded due to their limited ability to infer causality and generalize findings to clinical settings. By focusing on RCT-based meta-analyses, the study aimed to provide reliable insights into the effects of probiotics on gastrointestinal disorders.

Methodological quality assessment and data extraction

Two independent reviewers (MJ, PJ) assessed the methodological quality of the included articles using the Assessing the Methodological Quality of Systematic Reviews 2 (AMSTAR2) questionnaire [50]. The quality of the included meta-analyses was assessed using

the AMSTAR2 tool, a validated and widely recognized instrument for evaluating systematic reviews and meta-analyses. This tool examines 16 key items related to methodological rigor. These items include whether study selection and data extraction were performed independently by at least two reviewers to minimize bias, whether individual studies were evaluated for risk of bias using recognized tools such as the Cochrane Risk of Bias Tool, and whether publication bias was assessed using statistical methods like Egger's and Begg's tests or visual aids such as funnel plots. Additionally, the tool evaluates how heterogeneity among studies was analyzed, including the use of metrics like I^2 and Cochrane's Q -test, and whether explanations for observed variability were provided. The appropriateness of statistical methods, such as the use of fixed or random-effects models based on the characteristics of the pooled data, was also reviewed.

Each item in AMSTAR2 was graded as "Yes", "Partial Yes", or "No", and the overall quality of each meta-analysis was categorized as high, moderate, low, or critically low according to scoring guidelines. The detailed results of this assessment are summarized in Table 2. By employing this comprehensive quality evaluation framework, the study ensured that the included meta-analyses met high methodological standards, thereby enhancing the reliability and validity of the findings. Two reviewers (QZ, PL) independently also extracted the following data from the included meta-analyses: year of publication, sample size, study location, type of probiotics, mean dose and mean duration of studies included in the meta-analyses, effect sizes (ESs) including risk ratio (RR) and odds ratio (OR), and corresponding confidence intervals (CIs), which were subsequently recorded in an Excel spreadsheet. Any disagreements among the authors were resolved through consultation with a third author.

Data synthesis and statistical analysis

To obtain accurate results, the OR and RR effect sizes were analyzed separately for each outcome of interest. A random-effects model using advanced statistical modeling techniques [51] was employed to estimate the pooled effect size and its corresponding 95% CI. To assess heterogeneity, the I^2 statistic and Cochrane's Q -test were utilized. Heterogeneity was considered substantial if the I^2 value exceeded 50% or if the p -value for the Q -test was <0.1 . Subgroup analyses were conducted based on predefined variables, such as the treatment duration, sample size, dosage, and probiotic type. Sensitivity analysis was performed to evaluate the impact of individual study's removal on the overall effect size. Begg's and Egger's tests [52, 53] and visual inspection of funnel plot were performed to assess publication bias. If publication bias was present, the trim and fill analysis—a method to

adjust for potential publication bias—was carried out. All statistical analyses were conducted using Stata version 16 (Stata Corporation, College Station, TX, US), and a p -value less than 0.05 was considered statistically significant.

Results

Literature review

Initially, a total of 380 articles were identified through a comprehensive database search. After the removal of duplicates, 258 studies remained. Subsequently, following the evaluation of titles and abstracts, 216 articles were excluded. Of the 42 studies that proceeded to full-text screening, 28 were further excluded. Ultimately, 15 studies comprising 50 data sets were deemed eligible for inclusion in the umbrella meta-analysis. The study selection process and exclusion rationale are depicted in a detailed flow diagram (Fig. 1).

Characteristics of the included meta-analyses

The characteristics of 15 meta-analyses are presented in Table 1. Among them, eight articles were conducted in China [26, 32, 34, 35, 38, 39, 46, 48], four in USA [33, 37, 42, 54], two in Poland [36, 43], one in Canada [55]. Also, 14 articles have examined the effect of probiotic supplementation on the risk and odds of diarrhea [26, 32–39, 42, 46, 48, 54, 56]. Moreover, eight articles have investigated probiotics' impact on the risk and odds of nausea [26, 34–36, 38, 39, 43, 46]. Ten articles have reported the effect of probiotics on the risk and odds of epigastric pain [26, 33–35, 38, 39, 43, 46, 48, 55]. Five articles have investigated the impact of probiotics on the risk and odds of bloating [26, 35, 39, 43, 55]. Moreover, six articles have examined the effect of probiotics on the risk and odds of taste disturbance [26, 32, 35, 39, 43, 55].

Methodological quality assessment

Table 2 presents the findings of the quality assessment of meta-analyses according to the AMSTAR2 questionnaire. Five studies obtained a high ranking in terms of quality [26, 32, 37, 38, 55], and ten studies were of low quality [33–36, 39, 42, 43, 46, 48, 54].

Effects of probiotic supplementation on diarrhea

Based on OR analysis, the findings indicated that the supplementation of probiotics did not significantly impact the odds of diarrhea (OR 1.04; 95% CI 0.34 to 1.73; $p=0.97$) (see Fig. 2A). Furthermore, a significant heterogeneity was noted among the studies ($I^2=95.3\%$, P -heterogeneity <0.001). The sensitivity analysis findings indicated that excluding the Gong et al. study changed the significance of the overall result.

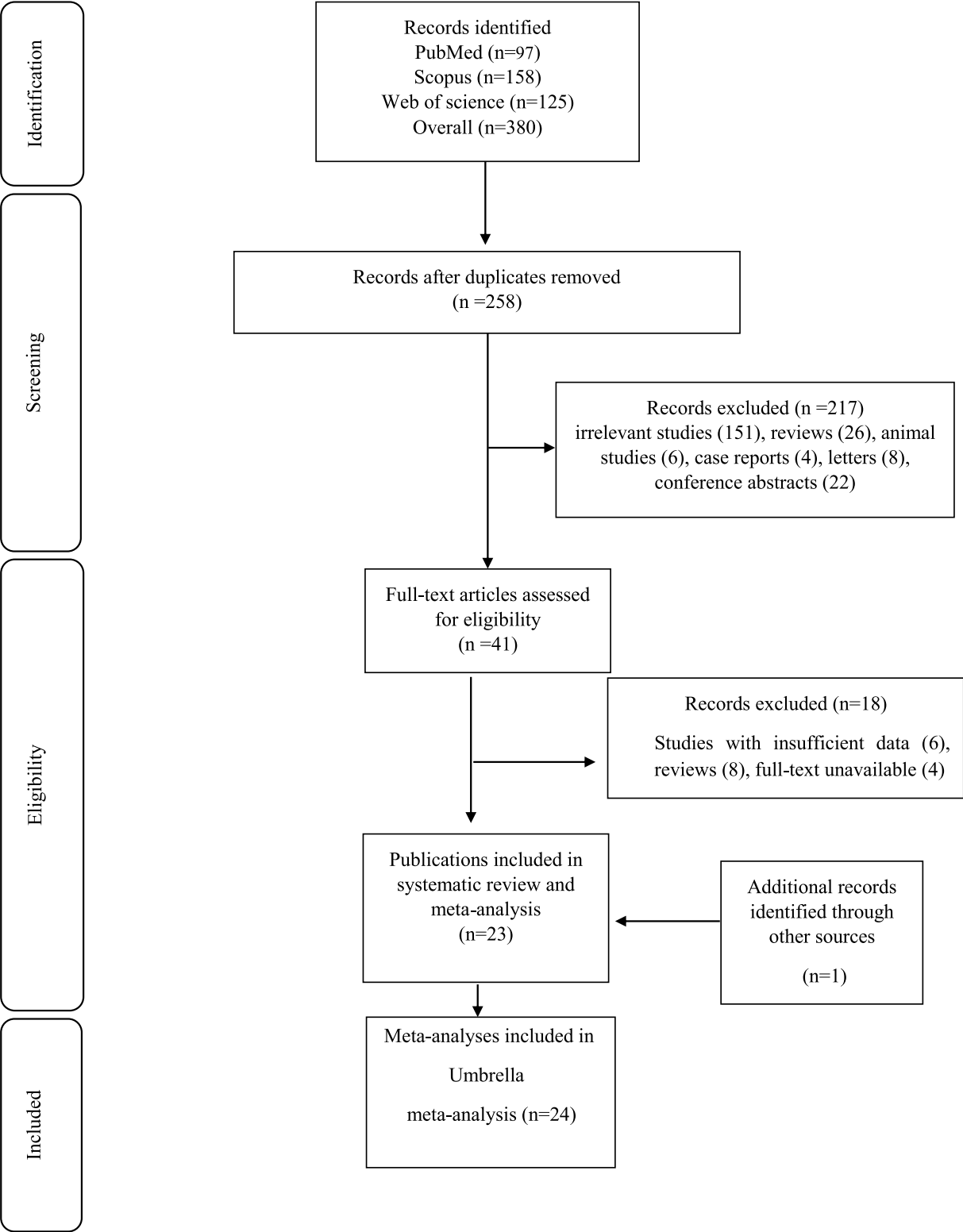


Fig. 1 PRISMA flow diagram of the study selection process

Table 1 Summary of characteristics of meta-analyses included in the umbrella review

First author	Year	Number of studies included	Type of probiotics	Health status	Probiotic dose (CFU/day)	Duration (week)	Primary gastrointestinal outcome	Risk of bias (included studies)	Heterogeneity
Goodoory	2023	32	Mixed	IBS	2.1×10^9	6	Epigastric pain	32 studies low risk	Significant
		26	Mixed		2.1×10^9	6	Bloating	26 studies low risk	Significant
Chenghai Yang	2021	6	<i>Lactobacillus</i>	<i>H. pylori</i> infections	10^8	10	Diarrhea	4 studies low risk, 1 study: moderate risk, 1 study: high risk	Non-significant
		6	<i>Lactobacillus</i>		10^8	10	Epigastric pain	4 studies low risk, 1 study: moderate risk, 1 study: high risk	Non-significant
Lynne V. McFarland	2021	15	Mixed	IBS	NR	NR	Epigastric pain	15 studies low risk of bias	Significant
Meiyi Zhang	2020	22	Mixed	<i>H. pylori</i> infections	NR	NR	Diarrhea	14 studies low risk, 8 studies moderate risk	Significant
		26	Mixed		NR	NR	Nausea	18 studies low risk, 8 studies moderate risk	Non-significant
		12	Mixed		NR	NR	Epigastric pain	NR	Non-significant
		8	Mixed		NR	NR	Bloating	NR	Non-significant
		10	Mixed		NR	NR	Taste disturbance	NR	Non-significant
		4	<i>Lactobacillus</i>		5.76×10^{10}	6	Diarrhea	4 studies low risk	Significant
Mingyang Yu	2019	3	<i>Lactobacillus</i>	<i>H. pylori</i> infections	2×10^9	6	Taste disturbance	3 studies low risk	Non-significant
		3	<i>Lactobacillus acidophilus</i> + <i>Bifidobacterium animalis</i>		14×10^{10}	1	Diarrhea	NR	Non-significant
McFarland	2016	4	<i>Lactobacillus acidophilus</i> + <i>Bifidobacterium bifidum</i>	<i>H. pylori</i> infections	3.5×10^9	1.5	Diarrhea	NR	Non-significant
		2	<i>Lactobacillus</i> + <i>Bifidobacterium</i> + <i>Streptococcus</i>		7.5×10^9	1.5	Diarrhea	NR	Non-significant
		2	<i>Lactobacillus acidophilus</i> , <i>Lactobacillus casei rhamnosus</i> , <i>Lactobacillus plantarum</i> , <i>Lactobacillus reuteri</i> , <i>Lactobacillus salivarius</i> , <i>Lactobacillus sporogenes</i> , <i>Bifidobacterium infantis</i> , <i>Bifidobacterium longum</i>		3×10^{10}	1	Diarrhea	NR	Non-significant
		2	<i>Lactobacillus acidophilus</i> , <i>Lactobacillus casei rhamnosus</i> , <i>Lactobacillus plantarum</i> , <i>Lactobacillus reuteri</i> , <i>Lactobacillus salivarius</i> , <i>Lactobacillus sporogenes</i> , <i>Bifidobacterium infantis</i> , <i>Bifidobacterium longum</i>		3×10^{10}	1	Diarrhea	NR	Non-significant

Table 1 (continued)

First author	Year	Number of studies included	Type of probiotics	Health status	Probiotic dose (CFU/day)	Duration (week)	Primary gastrointestinal outcome	Risk of bias (included studies)	Heterogeneity
Muhan LU	2016	9	<i>Lactobacillus</i> + <i>Bifidobacterium</i> + <i>Streptococcus</i>	<i>H. pylori</i> infections	71.5*10 ⁹	3	Diarrhea	9 studiES low risk	Significant
		5	<i>Lactobacillus</i> + <i>Bifidobacterium</i> + <i>Streptococcus</i>		83.4*10 ⁹	3	Nausea	5 studiES low risk	Non-significant
		6	<i>Lactobacillus</i> + <i>Bifidobacterium</i> + <i>Streptococcus</i>		69.6*10 ⁹	2.5	Epigastric pain	6 studiES low risk	Significant
Min-Min Zhang	2015	26	<i>Lactobacillus</i> + <i>Bifidobacterium</i> + <i>Streptococcus</i> + <i>Enterococcus</i> + <i>Saccharomyces</i> + <i>Clostridium</i>	<i>H. pylori</i> infections	NR	NR	Diarrhea	NR	Significant
		23	<i>Lactobacillus</i> + <i>Bifidobacterium</i> + <i>Streptococcus</i> + <i>Enterococcus</i> + <i>Saccharomyces</i> + <i>Clostridium</i>		NR	NR	Nausea	NR	Non-significant
		8	<i>Lactobacillus</i> + <i>Bifidobacterium</i> + <i>Streptococcus</i> + <i>Enterococcus</i> + <i>Saccharomyces</i> + <i>Clostridium</i>		NR	NR	Epigastric pain	NR	Non-significant
		13	<i>Lactobacillus</i> + <i>Bifidobacterium</i> + <i>Streptococcus</i> + <i>Enterococcus</i> + <i>Saccharomyces</i> + <i>Clostridium</i>		NR	NR	Bloating	NR	Non-significant
		19	<i>Lactobacillus</i> + <i>Bifidobacterium</i> + <i>Streptococcus</i> + <i>Enterococcus</i> + <i>Saccharomyces</i> + <i>Clostridium</i>		NR	NR	Taste disturbance	NR	Significant
Szajewska	2015	9	<i>Saccharomyces boulardii</i>	<i>H. pylori</i> infections	539 mg	2	Diarrhea	3 studiES low risk, 6 studies high risk	Non-significant
		6	<i>Saccharomyces boulardii</i>		625 mg	2.5	Nausea	3 studiES low risk, 1 study high risk, 2 studiES moderate risk	Non-significant

Table 1 (continued)

First author	Year	Number of studies included	Type of probiotics	Health status	Probiotic dose (CFU/day)	Duration (week)	Primary gastrointestinal outcome	Risk of bias (included studies)	Heterogeneity
McFarland	2015	9	<i>Saccharomyces boulardii</i>	<i>H. pylori</i> infections	1.3×10^{10}	2.5	Diarrhea	NR	Non-significant
		3	<i>Clostridium butyricum</i>		7×10^7	1.5	Diarrhea	NR	Non-significant
		5	<i>Lactobacillus rhamnosus</i>		9.6×10^9	2	Diarrhea	NR	Non-significant
		2	<i>Lactobacillus reuteri</i>		2.5×10^8	1.5	Diarrhea	NR	Non-significant
ZHIFA LV	2015	12	<i>Lactobacillus, Bifidobacterium, saccharo</i>	<i>H. pylori</i> infections	NR	3	Diarrhea	5 studies; high quality, 7 studies low quality	Significant
		10	<i>Lactobacillus, Bifidobacterium, Saccharomyces boulardii</i>		NR	3	Nausea	3 studies; high quality, 7 studies low quality	Significant
		8	<i>Lactobacillus, Bifidobacterium, Saccharomyces boulardii</i>		NR	2.5	Epigastric pain	4 studies; high quality, 4 studies low quality	Non-significant
Yi Gong	2015	19	<i>Lactobacillus, Bifidobacterium, Saccharomyces boulardii</i>	<i>H. pylori</i> infections	NR	8	Diarrhea	NR	Significant
		17	<i>Lactobacillus, Bifidobacterium, Saccharomyces boulardii</i>		NR	8	Nausea	NR	Non-significant
		9	<i>Lactobacillus, Bifidobacterium, Saccharomyces boulardii</i>		NR	8	Epigastric pain	NR	Non-significant
		10	<i>Lactobacillus, Bifidobacterium, Saccharomyces boulardii</i>		NR	8	Bloating	NR	Non-significant
		14	<i>Lactobacillus, Bifidobacterium, Saccharomyces boulardii</i>		NR	8	Taste disturbance	NR	Non-significant
Videlock	2012	24	<i>Lactobacillus + Bifidobacterium + Streptococcus</i>	Patients with diarrhea	NR	2	Diarrhea	NR	Non-significant
		10	<i>Lactobacillus + Bifidobacterium + Streptococcus</i>		NR	2	Diarrhea	NR	Non-significant

Table 1 (continued)

First author	Year	Number of studies included	Type of probiotics	Health status	Probiotic dose (CFU/day)	Duration (week)	Primary gastrointestinal outcome	Risk of bias (included studies)	Heterogeneity
Szajewska	2010	4	<i>Saccharomyces boulardii</i>	<i>H. pylori</i> infections	18.12*10 ⁹	2.5	Diarrhea	4 studiES low risk of bias	Non-significant
		3	<i>Saccharomyces boulardii</i>		17.5*10 ⁹	2.5	Nausea	3 studiES low risk of bias	Non-significant
		3	<i>Saccharomyces boulardii</i>		17.5*10 ⁹	2.5	Epigastric pain	3 studiES low risk of bias	Significant
		2	<i>Saccharomyces boulardii</i>		15*10 ⁹	2	Bloating	2 studiES low risk of bias	Non-significant
		4	<i>Saccharomyces boulardii</i>		18.12*10 ⁹	2.5	Taste disturbance	4 studiES low risk of bias	Non-significant
McFarland	2010	10	<i>Saccharomyces boulardii</i>	Patients' adult	11.35*10 ¹⁰	1.5	Diarrhea	5 studiES low risk, 2 study: moderate risk, 3 study: high risk	
TONG	2007	8	Different pro-biotic	<i>H. pylori</i> infections	7*10 ⁹	2	Diarrhea	NR	Non-significant
		7	Different pro-biotic		7*10 ⁹	2.5	Nausea	NR	Non-significant
		7	Different pro-biotic		7*10 ⁹	2.5	Epigastric pain	NR	Non-significant
		5	Different pro-biotic		7.3*10 ⁹	2.5	Taste disturbance	NR	significant

CFU: colony-forming units; *H. pylori*: *Helicobacter pylori*; IBS: irritable bowel syndrome; NR: not reported

Based on RR analysis, the findings derived from the analysis of 18 datasets indicated a significant decrease in the risk of diarrhea by 56% with the administration of probiotics supplements (RR 0.44; 95% CI 0.37 to 0.52; $p < 0.0001$) Fig. 2B. Furthermore, substantial heterogeneity was observed in the overall pooled analysis for diarrhea ($I^2 = 67.7\%$, $P < 0.001$), indicating variability among the included meta-analyses. Subgroup analyses for the outcome of diarrhea revealed that probiotic supplementation significantly reduced risk across all dosage categories, though heterogeneity was notably high in the $\geq 10^{10}$ group ($I^2 = 81.5\%$) and lower in others Table 3. A more pronounced effect was observed for interventions lasting ≤ 2 weeks, although substantial heterogeneity persisted ($I^2 = 76.7\%$). Regarding probiotic type, mixed strains yielded the strongest effect, but also demonstrated the highest heterogeneity ($I^2 = 91.7\%$). The effect was consistent across both high- and low-quality studies, with a slightly greater reduction seen in low-quality studies. Despite these findings, heterogeneity remained moderate to high in several subgroups, indicating that other unmeasured factors may contribute to the variability in outcomes. The sensitivity analysis indicated that excluding any individual study did not change the overall results. No significant small-study effect was seen

performing Egger's test ($p = 0.07$), like Begg's ($p = 0.93$). A visual examination of the funnel plot showed that the included studies were not evenly distributed. Thus, the trim and fill analysis was conducted. The findings continued showing significant decreasing impacts of probiotics on the risk of diarrhea by 59% (ES 0.41 95% CI 0.32 to 0.49) Fig. 2C.

Effect of probiotic supplementation on nausea

Based on OR analysis, the results of the study suggested that the administration of probiotics did not significantly impact the odds of nausea (OR 1.00; 95% CI 0.51 to 1.49; $p = 0.87$) (Fig. 3A). Moreover, the overall pooled analysis demonstrated significant heterogeneity ($I^2 = 90.3\%$, $P < 0.001$), suggesting considerable variation across the included meta-analyses. The sensitivity analysis demonstrated that excluding the study conducted by Gong et al. significantly changed the overall result.

Based on RR analysis, the findings from five articles indicated that the administration of probiotics led to a significant decrease in the risk of nausea by 41% (RR 0.59; 95% CI 0.49 to 0.60; $p < 0.001$) (see Fig. 3B). No considerable heterogeneity was observed ($I^2 = 0.00\%$, P -heterogeneity = 0.99). The sensitivity analysis demonstrated that

Table 2 Methodological quality assessment of included meta-analyses (based on AMSTAR 2)

First author	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	Q15	Q16	Overall
Goodoory	Yes	Yes	Yes	Partial yes	Yes	Yes	Yes	Partial yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	High quality
ZHIFA LV	Yes	Yes	Yes	Partial yes	Yes	Yes	Yes	Partial yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	High quality
Mingyang Yu	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	High quality
Meiyi Zhang	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Partial yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	High quality
McFarland 2015	Yes	Yes	Yes	Partial yes	Yes	Yes	Partial yes	Partial yes	Partial yes	No	Yes	Yes	Yes	Yes	Yes	Yes	High quality
Yang	Yes	Yes	Yes	Partial yes	Yes	Yes	Yes	Partial yes	Yes	No	Yes	Yes	No	Yes	Yes	Yes	Low quality
Yi Gong	Yes	Partial yes	Yes	Partial yes	Yes	Yes	Yes	Partial yes	Partial yes	No	Yes	Yes	No	No	Yes	Yes	Low quality
McFarland 2010	Yes	Yes	Yes	Partial yes	Yes	Yes	Yes	Partial yes	Partial yes	No	Yes	Yes	No	No	Yes	Yes	Low quality
McFarland 2016	Yes	Yes	Yes	Partial yes	Yes	Yes	Yes	Partial yes	Yes	No	Yes	Yes	No	Yes	Yes	Yes	Low quality
Muhan LU	Yes	Yes	Yes	Partial yes	Yes	Yes	Yes	Partial yes	Partial yes	No	Yes	Yes	No	Yes	Yes	Yes	Low quality
H. Szajewska 2010	Yes	Yes	Yes	Partial yes	Yes	Yes	Yes	Partial yes	Partial yes	No	Yes	Yes	No	Yes	Yes	Yes	Low quality
H. Szajewska 2011	Yes	Yes	Yes	Partial yes	Yes	Yes	Yes	Partial yes	Yes	No	Yes	Yes	No	Yes	Yes	Yes	Low quality
TONG	Yes	Yes	Yes	Partial yes	Yes	Yes	Yes	Partial yes	Yes	No	Yes	Yes	No	Yes	Yes	Yes	Low quality
Vidlock	Yes	Yes	Yes	Partial yes	No	Yes	Yes	Partial yes	Yes	No	Yes	No	No	Yes	Yes	Yes	Low quality
Min-Min Zhang	Yes	Yes	Yes	Partial yes	Yes	Yes	Yes	Partial yes	Yes	No	Yes	Yes	No	Yes	Yes	Yes	Low quality

Q1 (question)—Q16 correspond to AMSTAR 2 domains

Q1—Did the research questions and inclusion criteria for the review include the components of PICO?

Q2—Did the report of the review contain an explicit statement that the review methods were established prior to the conduct of the review, and did the report justify any significant deviations from the protocol?

Q3—Did the review authors explain their selection of the study designs for inclusion in the review?

Q4—Did the review authors use a comprehensive literature search strategy?

Q5—Did the review authors perform study selection in duplicate?

Q6—Did the review authors perform data extraction in duplicate?

Q7—Did the review authors provide a list of excluded studies and justify the exclusions?

Q8—Did the review authors describe the included studies in adequate detail?

Q9—Did the review authors use a satisfactory technique for assessing the risk of bias (RoB) in individual studies that were included in the review?

Q10—Did the review authors report on the sources of funding for the studies included in the review?

Q11—If meta-analysis was performed, did the review authors use appropriate methods for the statistical combination of results?

Q12—If a meta-analysis was performed, did the review authors assess the potential impact of RoB in individual studies on the results of the meta-analysis or other evidence synthesis?

Q13—Did the review authors account for RoB in individual studies when interpreting/discussing the review results?

Q14—Did the review authors provide a satisfactory explanation for and discussion of any heterogeneity observed in the review results?

Q15—If they performed quantitative synthesis, did the review authors conduct an adequate investigation of publication bias (small-study bias) and discuss its likely impact on the review results?

Q16—Did the review authors report any potential sources of conflict of interest, including any funding they received for conducting the review?

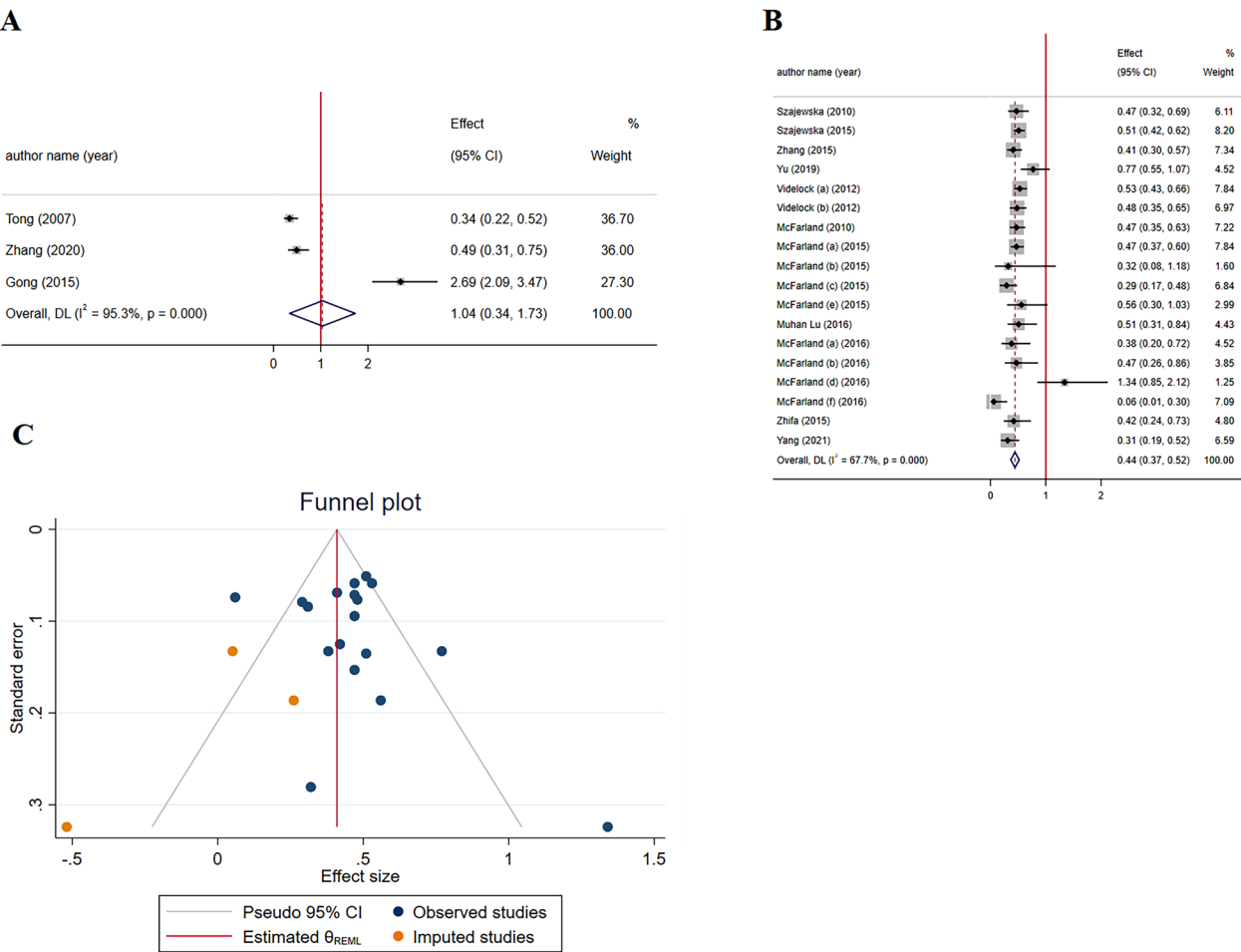


Fig. 2 The effects of probiotics on diarrhea: forest plot (A) detailing odds ratio (OR) and 95% confidence intervals (CIs); the effects of probiotics on diarrhea: forest plot (B) detailing risk ratio (RR) and 95% confidence intervals (CIs); and funnel plot for the effects of probiotics on diarrhea (C)

excluding any individual study did not change the overall result.

Effect of probiotic supplementation on epigastric pain

Based on OR analysis, the results showed that the probiotics supplementation did not affect the odds of epigastric pain (OR 0.85; 95% CI 0.48 to 1.24; $p = 0.42$) (Fig. 3C). Furthermore, the overall pooled analysis revealed a high degree of statistical heterogeneity ($I^2 = 71.7\%$, $P = 0.02$), reflecting substantial variability in the results reported across the included meta-analyses. The sensitivity analysis result showed that after excluding Gong's et al. study, the overall result became significant.

Based on RR analysis, the results obtained from 7 articles showed that probiotics supplementation significantly reduced the risk of epigastric pain by 29% (RR 0.71; 95% CI 0.56 to 0.87; $p < 0.0001$) (Fig. 3D). A notable level of heterogeneity was observed among

studies ($I^2 = 65.2\%$, P -heterogeneity = 0.008) underscoring significant discrepancies in effect estimates among the included meta-analyses. The sensitivity analysis showed that excluding any of the studies did not change the overall result. Subgroup analyses showed that probiotic supplementation at doses $< 10^{10}$ CFU and for durations < 4 weeks was significantly effective in reducing epigastric pain, with no observed heterogeneity in these subgroups ($I^2 = 0.0\%$). Similarly, significant effects were found in studies with durations ≥ 4 weeks and among high-quality studies, both also reporting no heterogeneity ($I^2 = 0.0\%$). In contrast, low-quality studies showed a nonsignificant effect with substantial heterogeneity ($I^2 = 75.2\%$). Although heterogeneity was generally low within effective subgroups, considerable heterogeneity remained in non-significant or unclassified groups, indicating possible influence from methodological or clinical variability (Table 3).

Table 3 Subgroup analyses of the effect of probiotic supplementation on gastrointestinal outcomes

	Number of studies	ES (95% CI)	P-within	I ² (%)	P-heterogeneity
Effects of probiotics on diarrhea					
Overall	18	0.44 (0.37, 0.52)	< 0.001	67.7	< 0.001
Dose					
≥ 10 ¹⁰	7	0.44 (0.28, 0.59)	< 0.001	81.5	< 0.001
NR	5	0.49 (0.43, 0.55)	< 0.001	0.00	0.68
< 10 ¹⁰	6	0.44 (0.26, 0.62)	< 0.001	58.2	0.03
Duration (week)					
> 2	6	0.47 (0.37, 0.58)	< 0.001	43.9	0.11
≤ 2	11	0.43 (0.31, 0.55)	< 0.001	76.7	< 0.001
NR	1	0.47 (0.21, 0.55)	< 0.001	0.00	
Study population					
Adult and children	8	0.46 (0.39, 0.54)	< 0.001	40.9	0.10
Adult	5	0.46 (0.36, 0.55)	< 0.001	22.3	0.27
Pediatric	1	0.48 (0.33, 0.63)	< 0.001	0.00	
NR	4	0.46 (0.10, 0.83)	< 0.001	85.5	< 0.001
Type of probiotics					
<i>Saccharomyces boulardii</i>	4	0.49 (0.42, 0.55)	< 0.001	0.00	0.94
Mixed	2	0.24 (0.11, 0.58)	< 0.001	91.7	< 0.001
<i>Lactobacillus</i>	4	0.45 (0.24, 0.67)	< 0.001	74	0.009
Lacto + Bifido + Strep	3	0.51 (0.43, 0.60)	< 0.001	0.00	0.87
Other	2	0.40 (0.18, 0.63)	< 0.001	0.00	0.74
<i>Lactobacillus</i> , <i>Bifidobacterium</i>	3	0.63 (0.22, 1.03)	0.069	71.6	0.02
Quality					
High	6	0.47 (0.33, 0.60)	< 0.001	53	0.059
Low	12	0.43 (0.34, 0.53)	< 0.001	73.7	< 0.001
Effects of probiotics on epigastric pain					
Overall	7	0.71 (0.56, 0.87)	< 0.001	65.2	0.008
Dose					
≥ 10 ¹⁰	2	0.64 (0.28, 1.01)	0.054	0.00	0.91
NR	3	0.92 (0.39, 1.44)	0.532	87.4	< 0.001
< 10 ¹⁰	2	0.72 (0.64, 0.80)	< 0.001	0.00	0.92
Duration (week)					
< 4	3	0.61 (0.37, 0.85)	< 0.001	0.00	0.96
NR	2	1.22 (0.12, 2.56)	0.764	93.6	< 0.001
≥ 4	2	0.72 (0.64, 0.80)	< 0.001	0.00	0.92
Quality					
High	2	0.71 (0.62, 0.80)	< 0.001	0.00	0.40
Low	5	0.79 (0.51, 1.08)	0.072	75.2	0.003

Effect size; CI confidence interval; I²: inconsistency across studies; P-within: P-value for within-group significance; P-heterogeneity: P-value for subgroup heterogeneity; NR: not reported

Effect of probiotic supplementation on bloating

Based on OR analysis, administering probiotics did not change the odds of bloating (OR 1.60; 95% CI 0.59 to 2.61; $p=0.65$) (Fig. 4A). Also, there was a notable level of heterogeneity among the studies ($I^2=90.3\%$, $P\text{-heterogeneity}=0.01$) reflecting substantial variability in the results reported across the included meta-analyses.

Based on RR analysis, three studies showed that taking probiotics supplements lowers the risk of bloating by 26% (RR 0.74; 95% CI 0.64 to 0.84; $P<0.001$) (Fig. 4B). There was no significant between-study heterogeneity ($I^2=0.00\%$, $P\text{-heterogeneity}=0.69$). The sensitivity analysis showed that the main results did not change if any of the studies were omitted.

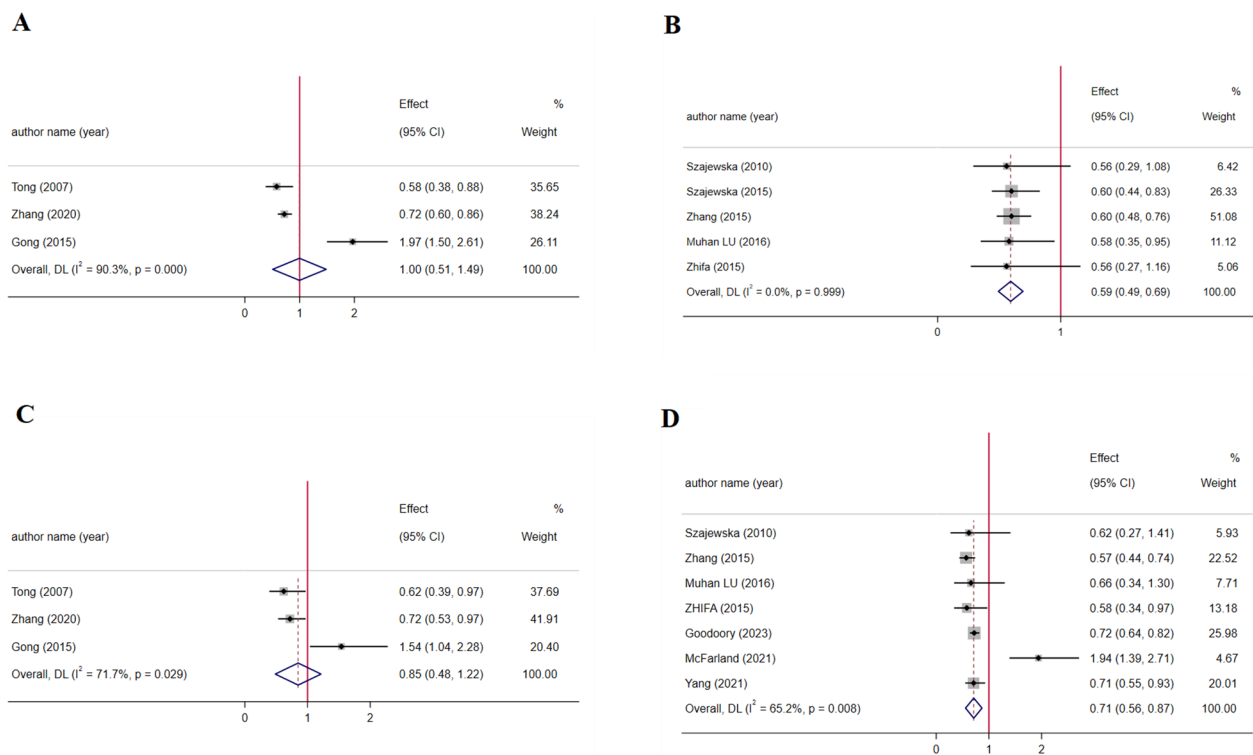


Fig. 3 The effects of probiotics on nausea: forest plot (A) detailing odds ratio (OR) and 95% confidence intervals (CIs); the effects of probiotics on nausea: forest plot (B) detailing risk ratio (RR) and 95% confidence intervals (CIs); the effects of probiotics on epigastric pain: forest plot (C) detailing odds ratio (OR) and 95% confidence intervals (CIs); the effects of probiotics on epigastric pain: forest plot (D) detailing risk ratio (RR) and 95% confidence intervals (CIs)

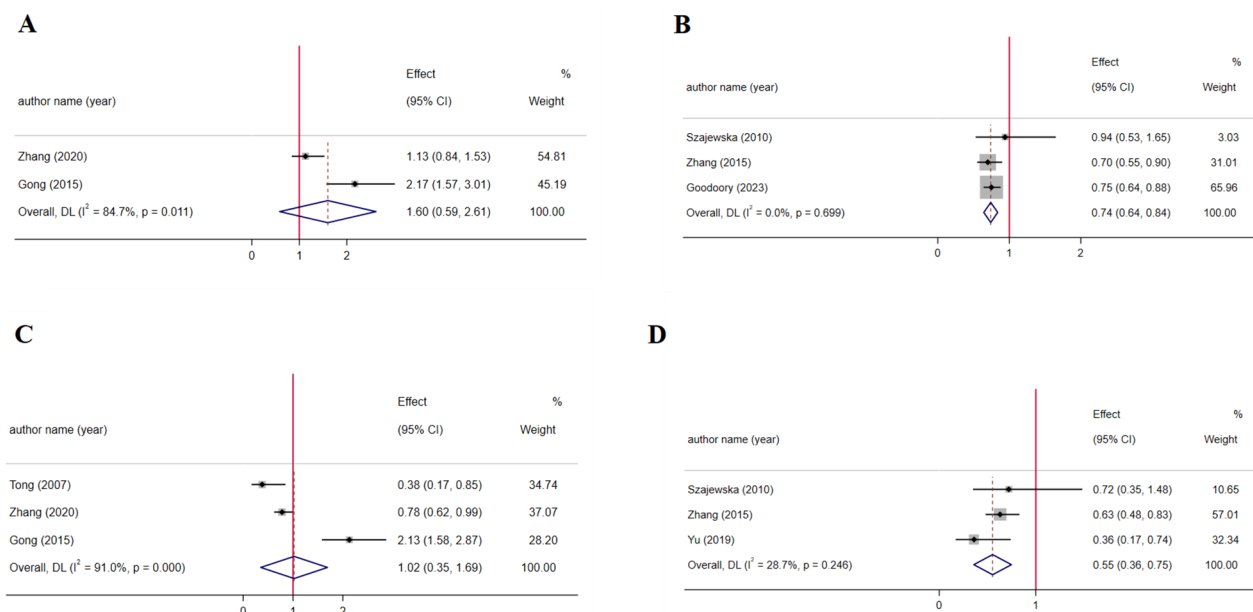


Fig. 4 The effects of probiotics on bloating: forest plot (A) detailing odds ratio (OR) and 95% confidence intervals (CIs); the effects of probiotics on bloating: Forest plot (B) detailing risk ratio (RR) and 95% confidence intervals (CIs); the effects of probiotics on taste disturbance: forest plot (C) detailing odds ratio (OR) and 95% confidence intervals (CIs); the effects of probiotics on taste disturbance: forest plot (D) detailing risk ratio (RR) and 95% confidence intervals (CIs)

Effect of probiotic supplementation on taste disturbance

Based on OR analysis, the findings demonstrated that probiotic use did not impact the odds of taste disturbance (OR 1.02; 95% CI 0.35 to 1.69; $p=0.95$; Fig. 4C). Additionally, there was significant heterogeneity among studies ($I^2=91.0\%$, P -heterogeneity <0.001) underscoring significant discrepancies in effect estimates among the included meta-analyses.

Based on RR analysis, probiotic supplementation significantly reduced the risk of taste disturbance by 45% based on data from three studies (RR 0.55; 95% CI 0.36 to 0.75; $P<0.001$) (Fig. 4D). Furthermore, there was no significant heterogeneity among studies ($I^2=28.7\%$, P -heterogeneity $=0.24$) suggesting considerable variation across the included meta-analyses. According to the findings of the sensitivity analysis, the overall result was changed by excluding the Gong s et al. study.

Discussion

This umbrella study analyzes 15 meta-analyses of clinical trials assessing the efficacy of probiotic supplementation in alleviating gastrointestinal disorders such as diarrhea, nausea, epigastric pain, bloating, and taste disturbances. The findings indicate that probiotics significantly reduce the risk of diarrhea nausea, epigastric pain, bloating, and taste disturbances.

Probiotics help mitigate gastrointestinal issues through several mechanisms. One involves competitive exclusion, where probiotics adhere to the gut lining and prevent pathogenic bacteria from colonizing, thereby reducing diarrhea and nausea. Notably, strains such as *Lactobacillus rhamnosus* GG and *Saccharomyces boulardii* have demonstrated this protective effect [57]. Another mechanism is the production of antimicrobial substances, including bacteriocins, organic acids, and hydrogen peroxide, which inhibit harmful bacteria like *Salmonella* and *Escherichia coli*, helping to regulate microbial activity in the gut [58]. Additionally, probiotics contribute to gut microbiota modulation by promoting beneficial bacteria while suppressing pathogenic ones. This balance, supported by strains such as *Lactobacillus casei* and *Lactobacillus rhamnosus*, reduces gastrointestinal disturbances, including diarrhea, nausea, and epigastric pain [59]. Furthermore, probiotics enhance gut barrier function by stimulating the development of tight junction proteins that maintain intestinal integrity. A stronger gut barrier prevents harmful substances from entering the bloodstream, reducing the likelihood of bloating, diarrhea, and epigastric pain. Studies have shown that *Lactobacillus plantarum* and *Bifidobacterium breve* effectively improve gut barrier function, further reinforcing the role of probiotics in gastrointestinal health. These

mechanisms collectively highlight the potential of probiotics in alleviating gastrointestinal disorder [60].

The reduction of bloating may be attributed to two principal mechanisms: reduced gas production and increased short-chain fatty acid (SCFA) generation. Probiotic strains like *Bifidobacterium lactis* and *Lactobacillus plantarum* help break down dietary carbohydrates more efficiently, leading to less gas formation and bloating relief. Additionally, probiotics ferment dietary fibers, producing SCFAs such as butyrate, propionate, and acetate, which support gut health and regulate gastrointestinal motility. By enhancing SCFA production, probiotics may help maintain gut motility, further reducing bloating [61, 62]. Regarding epigastric pain, in addition to the mentioned gut microbiota modulatory effect, the anti-inflammatory mechanism of probiotics is also effective in reducing it. They can modulate immune responses and reduce the production of pro-inflammatory substances. Epigastric pain can be associated with stomach or upper gastrointestinal tract inflammation. By mitigating inflammation, probiotics help alleviate epigastric pain in some instances [63]. Subgroup analyses indicated that multi-strain probiotics are more effective in reducing diarrhea risk than single-strain supplements. This advantage stems from a broader range of action and synergistic interactions between strains, enhancing survival, colonization, and intestinal activity. By promoting cross-feeding and cooperation, multi-strain probiotics improve gut microbiota modulation, strengthen the gut barrier, and enhance immune response, further reducing the risk of diarrhea [64]. Different probiotic strains influence various aspects of gut health, making multi-strain formulations more effective in addressing diarrhea's causes and progression. Some strains excel in pathogen inhibition through competitive exclusion, while others enhance antimicrobial production or immune modulation. Combining multiple strains ensures a comprehensive approach, maximizing efficacy in reducing diarrhea risk [65]. Also, the results of the subgroup analysis indicated that the duration of supplementation of <4 weeks has a more significant effect on reducing the risk of epigastric pain, which may reflect the prompt effect of probiotics in alleviating acute gastrointestinal symptoms [59].

Additionally, consuming multi-strain probiotics simultaneously can increase the likelihood of successful colonization and persistence in the gut. Different strains may have varying abilities to survive the harsh conditions of the gastrointestinal tract, adhere to intestinal epithelial cells, and establish a presence within the gut microbiota. Consuming a mixture of strains enhances the chances of sufficient colonization, crucial for exerting beneficial effects and reducing the risk of diarrhea [66, 67]. *Saccharomyces boulardii*, as shown in clinical studies [68],

is resistant to many commonly used antibiotics, allowing it to maintain its effectiveness in restoring microbial balance during antibiotic therapy. The mentioned features make the use of *Saccharomyces boulardii* strain bring better results. It is also noteworthy that the results of McFarland (d) 2016, McFarland (d) 2015, and Yu 2019 studies [32, 33, 37] have been insignificant, unlike the general result, which the possible reason could be due to the low number of included studies compared to others. Also, regarding diarrhea, nausea, bloating, and epigastric pain outcomes, several studies have obtained insignificant results, possibly due to the small sample size compared to other studies and similar to the previous outcome [32, 33, 37, 38, 43]. Regarding McFarland's study [37] about diarrhea outcome, in addition to the low sample size, the low average dose received can also be another reason for the inconsistency in the observed results. In the studies included to analyze the effect of probiotics on the risk of gastrointestinal bloating, the result of the Szajewska study [43], unlike the others, was insignificant, the possible reason for which was the use of a single strain as an auxiliary supplement, while the rest of the studies used a combination of several strains. A similar pattern was observed in the results of the analysis of epigastric pains. For instance, Szajewska [56] and Muhan LU [34] reported non-significant outcomes, possibly due to the use of single-strain probiotics rather than multi-strain formulations shown to be more effective in other studies [35, 38, 54, 55]. It is important to highlight that the RR, closely reflecting actual risk, is more suitable for clinical settings, aiding treatment decisions and risk assessments. In contrast, the OR is valuable in research, particularly for rare diseases, but may overestimate risks in common outcomes. While RR allows direct risk comparison, OR, based on odds, can exaggerate associations, potentially misleading interpretations. Clear distinction between the two is crucial for accurate communication with both clinicians and the public.

Strength and limitation

To the best of our knowledge, this study is one of the first umbrella meta-analyses to evaluate the effects of probiotics on gastrointestinal disorders. We conducted a comprehensive review of meta-analyses in this field, thoroughly assessing study quality. However, several limitations should be noted. A key limitation of this umbrella meta-analysis is the presence of heterogeneity across several pooled analyses. While subgroup analyses were conducted based on probiotic dose, duration, strain, and study quality—many of which reduced heterogeneity to negligible levels—some subgroups, particularly among low-quality studies or nonsignificant comparisons, still exhibited substantial heterogeneity

(e.g., $I^2 > 75\%$). These findings suggest that heterogeneity may stem not only from clinical factors such as population characteristics or intervention protocols, but also from methodological variation including differences in outcome definitions, risk of bias, and reporting quality among the included meta-analyses. Although subgrouping clarified part of the variability, the persistence of unexplained heterogeneity in some areas limits the strength and generalizability of the conclusions. Furthermore, the limited number of studies made it difficult to identify heterogeneity sources for diarrhea, nausea, epigastric pain, and bloating, analyzed using OR. The number of studies addressing outcomes such as bloating and taste disturbances was limited. Another limitation is that none of the included trials involved patients from UK or Africa. Finally, the study protocol was not registered in an accredited database, which may impact transparency.

Conclusion

Supplementation with probiotics can significantly reduce the risk of diarrhea, nausea, epigastric pain, bloating, and taste disturbance. Also, probiotics supplementation had a more robust effect on reducing the risk of epigastric pain at doses $> 10^{10}$ CFU. It also is more efficient in reducing the risk of diarrhea with supplementation < 2 weeks. Our findings support the beneficial role of probiotics in managing gastrointestinal symptoms, particularly diarrhea and epigastric pain. Subgroup analyses by dosage, duration, and strain strengthen these results. However, the presence of substantial heterogeneity and the predominance of low-quality meta-analyses limit the certainty of conclusions and highlight the need for more rigorous future research.

Data availability statement

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Author contributions

Q.Z: writing original draft, data curation, investigation; P.L: writing original draft, data curation, investigation, methodology, data analysis; H.W: writing original draft, data curation, investigation; Y.Z: writing original draft, data curation, investigation; Y.Z: writing original draft, data curation, investigation; S.A: writing original draft, data curation, investigation; P.J: writing original draft, data curation, investigation; M.J: writing review and editing; supervision, project administration, study design; H.U: writing review and editing; supervision, project administration, study design.

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Competing interests

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